

**EMAG Maschinenfabrik GmbH**

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EMAG Maschinenfabrik GmbH - Postfach 1152/62 - D-73080 Salach

Hyundai WIA Corp.
Mr. Tae Hyo Kim
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KOREA, REPUBLIC OF

Offer

Offers- No: 145205 / 3
date: 17/02/2015

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Your Reference

Inquiry dated 28/09/2014
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EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 250TWIN KBG for straight ball grooves (AC- joints)

for the simultaneous hard milling and turning of 2 components
of the same type

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Price does not include options (*)

Offer : 145205 / 3

Hyundai WIA Corp.

17/02/2015

2/29

Geschäftsführer:
Dr. Guido Hegener, Bernhard Böttigheimer,
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Handelsregister: Ulm HRB 533796
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Summary

1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 250TWIN KBG

1.1 BASIC MACHINE

1.1.17	1	PRICE OF MACHINE VSC250TWIN KBG (SIEMENS)	524.600,00 €
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1.2 ADDITIONAL EQUIPMENT

1.2.2	1	TOOL BLOCK FOR 2 TURNING TOOLS manually clamped	27.100,00 €
1.2.4	1	LINEAR LOADER (TWIN)	33.300,00 €
1.2.5	1	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK	1.100,00 €
1.2.6	1	WORKPIECE UNLOADING by motorised, chain-type conveyor	13.300,00 €
1.2.7	1	PART PUSHER FOR FINISH PARTS	2.250,00 €
1.2.8	3	ADDITIONAL SET OF JAW FOR PART PUSHER	450,00 €
1.2.9	1	SAFETY ENCLOSURE FOR THROUGH FEED CONVEYOR (Shuttle)	5.250,00 €
1.2.11	1	POSITIONING AND TRANSFER UNIT	30.600,00 €
1.2.12	3	ADDITIONAL SET OF JAWS FOR POSITIONING AND TRANSFER UNIT (SPARE)	2.850,00 €
1.2.13	1	BALL TRACK DETECTION SYSTEM BY LASER SENSOR	3.100,00 €
1.2.14	2	RIGID WORKPIECE RECEPTOR FOR OUTER RACE (21020-D3501)	3.400,00 €
1.2.15	2	RIGID WORKPIECE RECEPTOR FOR OUTER RACE (21022-0028)	3.400,00 €
1.2.16	2	RIGID WORKPIECE RECEPTOR FOR OUTER RACE (21024-0040A)	3.400,00 €
1.2.17	2	RIGID WORKPIECE RECEPTOR FOR OUTER RACE (21025-D3101)	3.400,00 €
1.2.19	1	SCRAPER-TYPE SWARF CONVEYOR	9.700,00 €
1.2.20	1	CHIP BOX	2.350,00 €
1.2.21	1	DRY-DUST AND MIST EXTRACTOR	4.700,00 €
1.2.22	1	AIR GUN	950,00 €
1.2.24	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS	5.550,00 €
1.2.25	1	WORKSHOP-ORIENTATED PROGRAMMING UF-JOINTS	5.550,00 €
1.2.26	1	EMAG REMOTE EXPERTS	2.650,00 €
1.2.27	1	NETWORK DRIVE CONNECTION	1.850,00 €
1.2.28	1	INTERFACE TO EXTERNAL HANDLING SYSTEM (RH), SIEMENS	3.350,00 €
1.2.30	1	AUXILIARY COOLING SYSTEM	5.650,00 €

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1.2.31	1	3-COLOUR STATUS LIGHT	1.000,00 €
1.2.32	1	SPECIAL MACHINE COLOUR	3.000,00 €
1.2.33	1	SPARE PARTS PACKAGE	40.000,00 €
1.2.34	1	MILLING SPINDLE (SPARE PART)	26.100,00 €
1.2.35	1	SPECIFIC CUSTOMER REQUIREMENTS	10.500,00 €

1.3 COMPONENT RELATED EQUIPMENT

1.3.2	1	SET OF TURNING TOOLS FOR HARD TURNING FOR OUTER RACE	22.600,00 €
1.3.3	16	SET OF MILLING TOOLHOLDERS HSK 63-C	16.000,00 €
1.3.4	1	SET OF MILLING TOOLS FOR HARD MILLING FOR OUTER RACE 21020-D3501	15.100,00 €
1.3.5	1	SET OF MILLING TOOLS FOR HARD MILLING FOR OUTER RACE 21022-0028	15.100,00 €
1.3.6	1	SET OF MILLING TOOLS FOR HARD MILLING FOR OUTER RACE 21024-0040A	11.400,00 €
1.3.7	1	SET OF MILLING TOOLS FOR HARD MILLING FOR OUTER RACE 21025-D3101	15.100,00 €
1.3.8	1	TEST BAR HSK 63-C	900,00 €
1.3.9	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE	1.800,00 €
1.3.11	2	ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER	7.950,00 €
1.3.12	2	WORKPIECE POSITION MONITORING, (AIR SENSING)	10.200,00 €
1.3.13	2	COLLET CHUCK 120	21.400,00 €
1.3.14	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120	6.800,00 €
1.3.15	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120	6.800,00 €
1.3.16	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120	6.800,00 €
1.3.17	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120	6.800,00 €
1.3.18	1	CHANGING FIXTURE FOR COLLET CHUCK 120	1.400,00 €
1.3.19	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (SPARE)	5.600,00 €
1.3.20	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (SPARE)	5.600,00 €
1.3.21	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (SPARE)	5.600,00 €
1.3.22	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (SPARE)	5.600,00 €

1.4 SERVICES

1.4.2	1	TOOLING LAYOUT FOR OUTER RACE 21020-D3501	5.600,00 €
1.4.3	1	TOOLING LAYOUT FOR OUTER RACE 21022-0028	4.200,00 €
1.4.4	1	TOOLING LAYOUT FOR OUTER RACE 21024-0040A	4.200,00 €
1.4.5	1	TOOLING LAYOUT FOR OUTER RACE 21025-D3101	4.200,00 €

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1.4.6	1	MACHINE SETUP FOR OUTER RACE 21020-D3501	6.950,00 €
1.4.7	1	MACHINE SETUP FOR OUTER RACE 21022-0028	5.250,00 €
1.4.8	1	MACHINE SETUP FOR OUTER RACE 21024-0040A	5.250,00 €
1.4.9	1	MACHINE SETUP FOR OUTER RACE 21025-D3101	5.250,00 €
1.4.10	1	PRE-ACCEPTANCE OF THE MACHINE	8.200,00 €
1.4.11	1	PRE-ACCEPTANCE OF THE MACHINE	6.150,00 €
1.4.12	1	FUNCTION TEST OF THE MACHINE	2.550,00 €
1.4.13	1	FUNCTION TEST OF THE MACHINE	2.550,00 €
1.4.14	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT	4.900,00 €
1.4.15	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT	4.900,00 €
1.4.16	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT	4.900,00 €
1.4.17	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT	4.900,00 €
1.4.18	1	PRODUCTION SUPPORT	16.000,00 €
1.4.19	1	INSTALLATION AND DEMONSTRATION (OVERSEAS)	31.300,00 €
1.4.23	3	ADDITIONAL SET OF DOCUMENTATION (INSTRUCTION MANUAL)	2.800,00 €
1.4.24	2	DOCUMENTATION ON CD	450,00 €
1.4.26	1	PACKING AND TRANSPORTATION OF TEST PIECE	1.800,00 €
1.4.27	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGE	950,00 €
1.4.28	1	FOB DELIVERY FREE GERMAN SEAPORT	5.700,00 €
1.4.29	1	SEAWORTHY PACKING	11.200,00 €

1.5 MACHINE SPECIFICATIONS

Total: **1.119.100,00 €**
Final negotiated price: **833.000,00 €**

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1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 250TWIN KBG

1.1 BASIC MACHINE

- 1.1.1 1 MACHINE BED
The swivelling axis with housing for the milling spindle is integrated in the machine bed. The machine bed manufactured from high quality Mineralit (cast granite) guarantees superior thermal stability and excellent dampening characteristics. Mineralit has excellent vibration dampening properties which are six to eight times better than cast iron. The large distance between the two high precision pre-stressed linear roller guideways of the X-axis guarantees high accuracy and high dynamics. The guideways are outside the working area and therefore require no protective covers.
- 1.1.2 1 MACHINING AREA
The machining area is effectively separated from the load/unload area by a partition wall facing the pick-up station.
A large door at the front of the machine provides access to the machining area. The door features a safety glass window to DIN EN ISO 23125:2010 specification.
A inspection lamp is installed in the machining area.
- 1.1.3 1 MULTI-FUNCTIONAL CROSS-SLIDE
This unit is equipped with two separate axes (X and Z) per spindle. Both spindles are controlled independently by the CNC to achieve separate tool correction. High rapid feed enable short idle times by component load/unload and gauging.
The axes drives are quick reaction frequency controlled service free AC motors which drive the slides over high precision ballscrew spindles. The X- and Z-axis are equipped with completely enclosed linear measuring systems.
On activation of the emergency-off switch or in case of power failure a feed motor with integrated brake will stop the vertical axis.
- 1.1.4 2 MOTOR SPINDLES
Each spindle headstock with motor spindle.
The motor spindles are a high dynamic frequency controlled service free AC asynchronous spindle motors.
High degree of spindle rigidity due to triple bearing support over a short distance. Precision shoulder bearings arranged in tandem (double-O). Additional bearing at spindle end for length compensation. All bearings are lifetime lubricated and thus maintenance free.
Sealing air to prevent dirt from penetrating from the outside.
The guideway system in the Z axis is equipped with hydrostatic bearings for compensation the cutting forces which guarantee excellent guideway quality by minimum friction.
Thermo-symmetrical design and integrated coolant system guarantee constantly maintained accuracy.
Positioning accuracy of spindle: +/- 0,01 angular degrees (C-axis)
With electro-hydraulic spindle locking.

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- 1.1.5 1 SWIVEL UNIT WITH MOTOR SPINDLE
FOR STRAIGHT GROOVES
Housing running in cylindrical roller bearings supported in the sidewalls of the machine base.
Swivel angle -30° to +70°
Driven by torque drive. Direct position feedback system.
- 2 MOTOR SPINDLES
- | | | |
|---------------------------------|-------|-------|
| Max. spindle speed | rpm | 6.000 |
| Bearing dia. | mm | 70 |
| Power rating at 100% duty cycle | kW | 15 |
| Full power at speed of | rpm | 3.000 |
| Tool register | HSK-C | 63 |
- Fluid-cooled spindle housing. Precision hybrid ceramic bearings. In-built thermistor motor protection. Rotating components precision balanced on two planes. Tool holder system with locating face on shoulder. Torsion-proof. Tools mechanically clamped.
(Mapal system)
(Motor spindles: warranty period 12 months)
- 1.1.6 1 ENERGY CONTAINER WITH CONTROL CABINET
Equipped to VDE 0113 and EN 60204, part 1 and integrated in the over-all machine frame, thus ensuring a totally transportable unit. 400 V machine voltage, 230 V control voltage (AC)/24 V DC.
The power supplies of the cross slides and main spindles are totally outside the machining and swarf area. Easy access to the axis drives, recirculating ballscrews etc. is possible through covers in the side walls or from the top of the machine.
- 1.1.7 1 HYDRAULIC UNIT
The hydraulic unit is equipped with a self regulating pump.
- 1.1.8 1 CENTRAL GREASE LUBRICATION SUPPLY SYSTEM
for lubrication of the recirculating ballscrews and guideway systems.
- 1.1.9 1 SERVICE UNIT FOR COMPRESSED AIR SUPPLY
- 1.1.10 1 COOLING UNIT (water - air)
A cooling unit controls the thermal stability of the machine. Spindle bearings, drive unit, motor for milling spindle and machine base are maintained at ambient temperature (10°C - 42°C). The electrical cabinet is cooled by 20°C.
- 1.1.11 1 CNC-CONTROL SIEMENS
According to control specification.
- 1.1.12 1 EMAG OPERATING AND SETTING MASKS
The EMAG masks offer the setter/ operator optimal support in setting and operating the machine.
Multiple base- and monitoring functions including data input are, depending on the machine type, available.

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- 1.1.13 1 **EMAG - REMOTE DIAGNOSIS - SIEMENS**
Control system: Sinumerik 840Dsl
- Linkage with user network necessary
- Connection via Internet and safe VPN technology to EMAG server
Scope of supply:
- Software for remote maintenance
- No additional equipment is supplied
Content:
- Remote access to NC / PLC / operating system for detailed error diagnosis and the possibility to support the customer in case of queries or problems.
Preconditions:
- Tele-diagnostics via Internet with VPN.
Your advantages:
- Quick, cost-effective support, resulting in reduced downtimes.
- Up-to-date problem solving without lengthy delays.
- Program extensions can be realised at a reduced cost.
- "On-the-spot" co-operation between specialist and operating/service personnel allows for complex errors to be diagnosed, analysed and eliminated quickly.
Use of the equipment past the warranty expiration date will be charged.
Option:
- UMTS-ROUTER
- EMAG REMOTE EXPERTS
- 1.1.14 1 For simple and fast programming of the machine on an appropriate ball track, the machine can be equipped with workshop-oriented programming modules. These modules are not included in the basic machine price and are described separately in the chapter „options to the CNC control“.
On the other hand, the part programs can also be completed by the operator in normal ISO code.
- 1.1.15 1 **STANDARD COLOUR**
Machine - light grey to RAL 7035,
front door - grey to RAL 7046,
automation, swarf conveyor - grey to RAL 7046.
- 1.1.16 1 **NOISE LEVEL**
A maximum sound level of 78 dB(A) to DIN 45635 is obtained when turning with a continuous cut on a C45 steel disc in both longitudinal and transverse direction. If the workpiece has for example an interrupted cut due to its geometry or its clamping condition there may occur a higher sound level.
For an extra cost sound deadening material can be fitted to reduce this higher sound level.
- 1.1.17 1 **PRICE OF MACHINE VSC250TWIN KBG (SIEMENS)** 524.600,00 €

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1.2 ADDITIONAL EQUIPMENT**1.2.1 TOOLING SYSTEM**

- 1.2.2 1 TOOL BLOCK FOR 2 TURNING TOOLS manually clamped 27.100,00 €
consisting of:
cylindrical shaft toolholder for Capto C6.
With internal air blast feeding to the tools. The coolant system guarantees constantly maintained accuracy.

1.2.3 COMPONENT STORAGE AND TRANSPORT SYSTEM

- 1.2.4 1 LINEAR LOADER (TWIN) 33.300,00 €
Linear conveyor from which the spindle collects the workpieces.
With cardanic, spring-loaded pick-up station, for the positive location of workpieces in chuck and cassettes used to convey the workpieces.
Prepared for fitting of an automatic alignment device for the correct angular positioning of workpieces in the ball tracks (hard machining).
CV joints are conveyed vertically, with the CV joint bell pointing downward.
- 1.2.5 1 AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK 1.100,00 €
For the cleaning of the pick-up unit and the clamping chuck by air blast. The air is processed by a service unit and directed at the component through 2 nozzles.
- 1.2.6 1 WORKPIECE UNLOADING by motorised, chain-type conveyor 13.300,00 €
Workpiece diameter: 40 - 200 mm
Length: approx. 1800 mm
Conveyors mounted to left of machine.
Includes a interface for customer provided finish part storage.
Manufacturer: Rexnord or MK
- 1.2.7 1 PART PUSHER FOR FINISH PARTS 2.250,00 €
pneumatically operated,
for to move the finish parts from the linear loader to the unloading conveyor
Mounted on the left side of the machine
- 1.2.8 3 ADDITIONAL SET OF JAW FOR PART PUSHER 450,00 €
For the pneumatic part pusher for finish parts.
- 1.2.9 1 SAFETY ENCLOSURE FOR THROUGH FEED CONVEYOR (Shuttle) 5.250,00 €
Enclosure covering moving automation components. 2 Access doors mechanically and electrically lockable.
Made as EMAG standard.

Price does not include options (*)

Offer : 145205 / 3

Hyundai WIA Corp.

17/02/2015

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Geschäftsführer:
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1.2.10 AUXILIARY EQUIPMENT FOR HARD MACHINING

1.2.11	1	POSITIONING AND TRANSFER UNIT Alignment device for angular positioning of the workpieces according to the ball tracks and following transfer to the workpiece receptor on the linear loader. The unit is equipped with two NC controlled linear axes and one rotation axis, also a three-jaw-gripper and one set of gripper jaws laid out for the workpiece sizes 20, 22, 23 and 24 (diameter between 60-62mm).	30.600,00 €
1.2.12	3	ADDITIONAL SET OF JAWS FOR POSITIONING AND TRANSFER UNIT (SPARE) One set of gripper jaws laid out for the workpiece sizes 20, 22, 24 and 25.	2.850,00 €
1.2.13	1	BALL TRACK DETECTION SYSTEM BY LASER SENSOR additional price for laser sensor instead of induction sensor.	3.100,00 €
1.2.14	2	RIGID WORKPIECE RECEPTOR (21020-D3501) for one type of Outer Race. Mounted on the linear loader.	3.400,00 €
1.2.15	2	RIGID WORKPIECE RECEPTOR (21022-0028) for one type of Outer Race. Mounted on the linear loader.	3.400,00 €
1.2.16	2	RIGID WORKPIECE RECEPTOR (21024-0040A) for one type of Outer Race. Mounted on the linear loader.	3.400,00 €
1.2.17	2	RIGID WORKPIECE RECEPTOR (21025-D3101) for one type of Outer Race. Mounted on the linear loader.	3.400,00 €
1.2.18		COOLANT AND SWarf DISPOSAL SYSTEM	
1.2.19	1	SCRAPER-TYPE SWarf CONVEYOR Swarf exit height 1100 mm. Located to the rear of the machine. Usage: for short swarf, in particular grey cast iron.	9.700,00 €
1.2.20	1	CHIP BOX with caster	2.350,00 €
1.2.21	1	DRY-DUST AND MIST EXTRACTOR For dry hard machining. Including stainless steel separator baffle. High-precision filtration unit with filter inserts. Air volume 1000 m3/h. Company REVEN	4.700,00 €
1.2.22	1	AIR GUN with hose	950,00 €

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1.2.23 OPTIONAL EQUIPMENT FOR CNC CONTROL

- | | | | |
|--------|---|---|------------|
| 1.2.24 | 1 | <p>WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS</p> <p>For outer races (straight grooves).
For quick and easy programming parameters for componentrelated dimensions are adapted.
Same process for tool-offset.
Every tool is allocated an empirically determined preset quantity of workpieces to be machined.
On expiry of the tool life the machine is stopped.
Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per tool.</p> | 5.550,00 € |
| 1.2.25 | 1 | <p>WORKSHOP-ORIENTATED PROGRAMMING UF-JOINTS</p> <p>For outer races (straight grooves).
For quick and easy programming parameters for componentrelated dimensions are adapted.
Same process for tool-offset.
Every tool is allocated an empirically determined preset quantity of workpieces to be machined.
On expiry of the tool life the machine is stopped.
Control monitor display: tool-no., preset and remaining tool life quantity of workpieces per tool.</p> | 5.550,00 € |
| 1.2.26 | 1 | <p>EMAG REMOTE EXPERTS</p> <p>Price per year 2.500,00 €
(EMAG Remote Diagnosis included)
CNC control Sinumerik 840Dsl with PCU50 and HMI-Advanced</p> <ul style="list-style-type: none">- Assistance key- 24/7 telephone support via our service hotline- Quarterly documentation on machine evaluations and breakdowns (reports on efficiency, reliability, connection monitoring, service requests, etc.)- If required, consultation with remote experts, expert service worldwide (Mo - Fr 0700h - 1700h GMT + 0100) | 2.650,00 € |
| 1.2.27 | 1 | <p>NETWORK DRIVE CONNECTION
(Control system Siemens 840Dsl)</p> <p>Linking the machine control to your network after startup in your factory, so that network drives are available in the user interface. NC-programs or any data can be stored or loaded from there.</p> <p>Customer-specific requirements / restrictions for network integration:</p> <ul style="list-style-type: none">- Operation with DHCP server yes / no; if no, then IP address, sub-net mask and default gateway- DNS server and enable code of network source (also IP-address of server, if DNS not known)- User name and password for the automatic preparation of the connection- Network cable into the electrical cabinet <p>When providing server and directory names it has to be taken into account that the 840Dsl features an English language Windows operating system that cannot handle the German umlauts ä, ö and ü.</p> <p>When installing of security update, the company Siemens restrictions are to consider. If in the domain automatically software is installed on all computers, so the functionality is no longer guaranteed after restart!</p> | 1.850,00 € |

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1.2.28	1	INTERFACE TO EXTERNAL HANDLING SYSTEM (RH), SIEMENS for customer provided loading conveyor	3.350,00 €
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1.2.29 OTHER AUXILIARY EQUIPMENT

1.2.30	1	AUXILIARY COOLING SYSTEM for ambient temperatures of > 40°C consisting of auxiliary coolant supply unit for switchgear cabinet and also for hydraulic unit.	5.650,00 €
1.2.31	1	3-COLOUR STATUS LIGHT mounted on machine. Colour code: red: machine stopped yellow: warning -tool to be changed, -lubricant running low, etc. green: machine in automatic cycle. Supplier is Q-Ligh (Korea)	1.000,00 €
1.2.32	1	SPECIAL MACHINE COLOUR Complete machine according to customers colour specification.	3.000,00 €
1.2.33	1	SPARE PARTS PACKAGE Budgetary amount for spare parts package (Preliminary, final definition and list of selected components will be forwarded in case of an order) This package is fixed priced, non discountable.	40.000,00 €
1.2.34	1	MILLING SPINDLE (SPARE PART) One milling sindle as spare part	26.100,00 €

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1.2.35 1 **SPECIFIC CUSTOMER REQUIREMENTS** 10.500,00 €
According to "WIA standard 2014 May 12", "Part Maker List" and "Specification of Quotation"

"General Specification" is **not** considered.

DEVIATIONS FROM THE "SPECIFICATIONS OF QUOTATION"

the following points can not be guaranteed or have to be clarified with customer:

- Page 1, "Condition of FAC Issue"
It's needed a definition of "alarm", what is included and what is not included?
- Page 1, "Productivity -> Efficiency"
Tool exchange is not considered
- Page 1, "Others -> Operation height"
Shuttle height is 1050mm, so loading conveyor has to be obliquely
- Page 1, "Surrounding -> Surrounding temp."
Temperature between 10°C - 45°C
- Page 2, "Submission of Documentation -> Quotation"
Tool list / life, Change over procedure with timing, DWG of Major parts, Schedule is not part of the quotation
- Page 3, "Specification -> Workpiece quality"
Ball diameter is no acceptance criteria
- Page 6, "Detail -> OLT & Pre-acceptance"
Burning, hardness test made by Hyundai WIA in Korea before Pre-acceptance
- Page 6, "Detail -> Training"
Training (operator and maintenance) is only for one shift
- Page 7, "Detail -> Training"
Training manual in English and Chinese version (see page 2, "Delivery")
- Page 8, "Detail -> Others"
Unloading conveyor from Tsubaki is not possible, EMAG standard is Rexnord or MK
- Page 9, "Detail -> Color"
Safety fence made as EMAG standard
- Page 11, "Detail -> Color"
Motor cover paint in RAL 7035, Mesh/ acryle cover is not necessary
- Page 12, "Electric -> Main circuit"
EMAG standard solenoid valve: 24V DC with LED
- Page 12, "Electric -> Main circuit"
Frequency of control voltage, solenoid valve, consent is also 50 Hz as main power
- Page 12, "Electric -> CNC & PLC"
EMAG standard voltage: 400V AC / 24V DC
- Page 12, "CNC & PLC"
EMAG standard interface: USB port instead of serial interface (RS 232C)
- Page 13, "Part -> Automation"
Infeed conveyor is not part of this quotation
- Page 14, "Unit -> Others -> Wiring"
Only 10% marginal space are possible

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1.3 COMPONENT RELATED EQUIPMENT**1.3.1 TOOL EQUIPMENT**

1.3.2	1	SET OF TURNING TOOLS FOR HARD TURNING FOR OUTER RACE consisting of: - 4 clamping units Capto C6 - 4 basic tool holder Capto C6 - 130 CBN cutting inserts	22.600,00 €
1.3.3	16	SET OF MILLING TOOLHOLDERS HSK 63-C 1 pcs toolholders for interchangeable milling heads	16.000,00 €
1.3.4	1	SET OF MILLING TOOLS FOR HARD MILLING FOR OUTER RACE 21020-D3501 Head system with 4 brazed CBN cutting inserts consisting of: - 40 cutting heads	15.100,00 €
1.3.5	1	SET OF MILLING TOOLS FOR HARD MILLING FOR OUTER RACE 21022-0028 Head system with 4 brazed CBN cutting inserts consisting of: - 40 cutting heads	15.100,00 €
1.3.6	1	SET OF MILLING TOOLS FOR HARD MILLING FOR OUTER RACE 21024-0040A Head system with 4 brazed CBN cutting inserts consisting of: - 30 cutting heads	11.400,00 €
1.3.7	1	SET OF MILLING TOOLS FOR HARD MILLING FOR OUTER RACE 21025-D3101 Head system with 4 brazed CBN cutting inserts consisting of: - 40 cutting heads	15.100,00 €
1.3.8	1	TEST BAR HSK 63-C For Geometry test of the milling spindles, including wooden box for storage	900,00 €
1.3.9	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE consisting of: - 1 cone wiper - 1 torque wrench with insert - 2 plugs HSK63 - 1 ball plug gauge HSK63	1.800,00 €

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1.3.10 CHUCKING SYSTEM

1.3.11	2	ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER for spindle nose 6. Max. drawbar force 2950 daN at 60 bar. The clamping pressure is manually infinitely adjustable and controlled by a pressure switch. The clamping cylinder stroke is controlled by a linear measuring system.	7.950,00 €
1.3.12	2	WORKPIECE POSITION MONITORING, (AIR SENSING) The position of the workpiece in the clamping fixture is monitored by measuring the air pressure between the part stop and the component. This establishes contact accuracy. Inaccuracies on the workpiece resulting from deformation of clamping fixture or workpiece etc., are not recognised. The air is supplied by one or more nozzles (depending on clamping fixture type). The pressure is measured with a pressure switch. The air supply is controlled by a directional valve. The air flow is cleaned by a service unit and sent through the main spindle. NOTE: Requires CHUCKING CYLINDER WITH AIR THROUGH THE SPINDLE. Equipping the clamping fixture with a pneumatic system monitor is not included in the price of this item.	10.200,00 €
1.3.13	2	COLLET CHUCK 120 Outer dia. 235 mm With basic stop. Clamping range 60 -120 mm. Prepared for air sensing. Without workpiece-specifically interchangeable parts.	21.400,00 €
1.3.14	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 Laid out for Outer Race 21020-D3501. Consisting of clamping head, part stop and control ring.	6.800,00 €
1.3.15	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 Laid out for Outer Race 21022-0028. Consisting of clamping head, part stop and control ring.	6.800,00 €
1.3.16	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 Laid out for Outer Race 21024-0040A. Consisting of clamping head, part stop and control ring.	6.800,00 €
1.3.17	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 Laid out for Outer Race 21025-D3101. Consisting of clamping head, part stop and control ring.	6.800,00 €
1.3.18	1	CHANGING FIXTURE FOR COLLET CHUCK 120 for quick changing of the clamping head. Pneumatically operated.	1.400,00 €

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1.3.19	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (SPARE) Laid out for Outer Race 21020-D3501. Consisting of clamping head and part stop.	5.600,00 €
1.3.20	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (SPARE) Laid out for Outer Race 21022-0028. Consisting of clamping head and part stop.	5.600,00 €
1.3.21	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (SPARE) Laid out for Outer Race 21024-0040A. Consisting of clamping head and part stop.	5.600,00 €
1.3.22	2	SET OF INTERCHANGEABLE PARTS FOR COLLET CHUCK 120 (SPARE) Laid out for Outer Race 21025-D3101. Consisting of clamping head and part stop.	5.600,00 €

Price does not include options (*)

Offer : 145205 / 3

Hyundai WIA Corp.

17/02/2015

17/29

Geschäftsführer:
Dr. Guido Hegener, Bernhard Böttigheimer,
Jürgen Müller, Norbert Heßbrüggen
Handelsregister: Ulm HRB 533796
Steuer-Nr.: 63002/12866 FA Göppingen
Ust-IdNr.: DE 813 898 840

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1.4 SERVICES

1.4.1	1	Please note that no additional services are contained in the previously specified prices. We would be pleased to carry out the following services and would quote according to your wishes. - Preparation of tooling layouts and NC programs - Set-up machine for component trials on your specific workpiece in our works - Machine approval in our works by your engineers. - Final acceptance at customer's plant with a specific workpiece	
1.4.2	1	TOOLING LAYOUT FOR OUTER RACE 21020-D3501 Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	5.600,00 €
1.4.3	1	TOOLING LAYOUT FOR OUTER RACE 21022-0028 Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	4.200,00 €
1.4.4	1	TOOLING LAYOUT FOR OUTER RACE 21024-0040A Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	4.200,00 €
1.4.5	1	TOOLING LAYOUT FOR OUTER RACE 21025-D3101 Generation of a customer-specific tooling layout with tool selection, NC workpiece program, clamping fixtures and attachments.	4.200,00 €
1.4.6	1	MACHINE SETUP FOR OUTER RACE 21020-D3501 Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	6.950,00 €
1.4.7	1	MACHINE SETUP FOR OUTER RACE 21022-0028 Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	5.250,00 €
1.4.8	1	MACHINE SETUP FOR OUTER RACE 21024-0040A Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	5.250,00 €
1.4.9	1	MACHINE SETUP FOR OUTER RACE 21025-D3101 Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	5.250,00 €

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1.4.10	1	PRE-ACCEPTANCE OF THE MACHINE Specific workpiece: Outer Race 21020-D3501 including measuring of workpieces at EMAG (3D measuring machine, FAG roundness measuring device, etc.). One workpiece in one clamping operation. Max. 50 components. Submission of proof to Cm of machine's capability: max. 3 measuring characteristics.	8.200,00 €
1.4.11	1	PRE-ACCEPTANCE OF THE MACHINE Specific workpiece: Outer Race 21022-0028 including measuring of workpieces at EMAG (3D measuring machine, FAG roundness measuring device, etc.). One workpiece in one clamping operation. Max. 50 components. Submission of proof to Cm of machine's capability: max. 3 measuring characteristics.	6.150,00 €
1.4.12	1	FUNCTION TEST OF THE MACHINE Specific workpiece: Outer Race 21024-0040A They will be machined 10 parts each for each spindle. All dimensions must be within their tolerance one hundred percent.	2.550,00 €
1.4.13	1	FUNCTION TEST OF THE MACHINE Specific workpiece: Outer Race 21025-D3101 They will be machined 10 parts each for each spindle. All dimensions must be within their tolerance one hundred percent.	2.550,00 €
1.4.14	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT Specific workpiece: Outer Race 21020-D3501 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 50 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test.	4.900,00 €

PLEASE NOTE

Pre-machining-conditions:

- Roundness of clamping diameter: 0.01-0.015mm
- Flatness of part stop: max. 0.015mm
- Rectangularity (OD to end stop): max. 0.015mm

Also we guarantee a machine capability only in a warm machine condition.

Price does not include options (*)

Offer : 145205 / 3

Hyundai WIA Corp.

17/02/2015

19/29

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1.4.15	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT Specific workpiece: Outer Race 21022-0028 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 50 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test. PLEASE NOTE Pre-machining-conditions: - Roundness of clamping diameter: 0.01-0.015mm - Flatness of part stop: max. 0.015mm - Rectangularity (OD to end stop): max. 0.015mm Also we guarantee a machine capability only in a warm machine condition.	4.900,00 €
1.4.16	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT Specific workpiece: Outer Race 21024-0040A Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 50 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test. PLEASE NOTE Pre-machining-conditions: - Roundness of clamping diameter: 0.01-0.015mm - Flatness of part stop: max. 0.015mm - Rectangularity (OD to end stop): max. 0.015mm Also we guarantee a machine capability only in a warm machine condition.	4.900,00 €
1.4.17	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT Specific workpiece: Outer Race 21025-D3101 Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 50 workpieces Evaluation and guaranteed criteria are the same as for pre-acceptance test. PLEASE NOTE Pre-machining-conditions: - Roundness of clamping diameter: 0.01-0.015mm - Flatness of part stop: max. 0.015mm - Rectangularity (OD to end stop): max. 0.015mm Also we guarantee a machine capability only in a warm machine condition.	4.900,00 €
1.4.18	1	PRODUCTION SUPPORT 2 weeks production support at customers plant Performed by EMAG China	16.000,00 €

Price does not include options (*)

Offer : 145205 / 3

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1.4.19 1 INSTALLATION AND DEMONSTRATION (OVERSEAS) 31.300,00 €
in your works

The following services are included:

- Machine installation and demonstration
- Machine approval and hand-over
- Talk-through of the machine documentation and service instruction
- Operator instruction
- Travel costs

Total of 15 man days

Not included are:

- Machine set-up
- Machine operation for production
- Additional service

Additional service engineer costs for time in excess of that quoted will be charged extra.

1.4.20 1 INSTALLATION POINT
The installation point has to be prepared in accordance with the documentation supplied. We take it for granted that all conditions are met before our installation crew arrives.
Where waiting times are incurred due to the conditions not having been met, these will be charged extra. During installation and commissioning our engineers have to be provided, on request, with the support of skilled customer personnel and similar assistance. This also applies to the operation of cranes, handling equipment and other aids.

1.4.21 1 MACHINE LIFTING EQUIPMENT
EMAG will provide a lifting equipment to lift the machine by crane, free of charge. However, should the customer not return the equipment to EMAG within 4 weeks, freight and charges prepaid, a separate invoice of EUR 1.500,00 will be issued.

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1.4.22 1 DOCUMENTATION (Siemens)

OPERATING INSTRUCTIONS

1 complete set of operating instructions on paper

Language: English

Manuals for the machine:

- Introduction
- Safety instructions
- Product description
- Transport and installation
- Start-up
- Production
- Preventive maintenance

Manuals from the manufacturer of the control unit:

Siemens SIN 840D sl

- User manual
- Diagnostics manual
- Programming - Basics
- Programming - Cycles
- Programming - Operations planning and scheduling

Machine diagrams:

- Machine layout
- Transport diagram
- Machine data sheet
- Drive data sheet
- Connection data sheet for fluidics
- Connection data sheet for electrics

List of spare parts and wearing parts

Operating instructions from third parties

Documentation for clamping devices and tools

TECHNICAL DATA

1 set of technical data on paper, German or English

- Electrical circuit diagram with list of el. devices
- Fluid system diagram with list of el. devices
- Drawings of subassemblies with relevance for spare parts
- List of components with relevance for spare parts and wearing parts in relation to the drawings
- Technical data, test reports, declarations of incorporation from third parties

Other languages available on request for an additional charge; third party documentation excepted

1.4.23	3	ADDITIONAL SET OF DOCUMENTATION (INSTRUCTION MANUAL) 1 additional set of the documentation described above. Language: Chinese	2.800,00 €
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1.4.24	2	DOCUMENTATION ON CD 1 complete set of documentation on CD	450,00 €
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1.4.25 TRANSPORT COSTS

1.4.26	1	PACKING AND TRANSPORTATION OF TEST PIECE Transportation from China	1.800,00 €
1.4.27	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGE Transportation from Korea	950,00 €
1.4.28	1	FOB DELIVERY FREE GERMAN SEAPORT	5.700,00 €
1.4.29	1	SEAWORTHY PACKING Conservation of the entire cargo and packaging in a wooden box	11.200,00 €

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1.5 MACHINE SPECIFICATIONS

1.5.1 1 TECHNICAL MACHINE DATA VSC250TWIN KBG Subject to change without notice

WORKING AREA

Chuck diameter	mm	250
Swing diameter	mm	260
X-stroke	mm	850
Z-stroke	mm	200

WORKPIECE DIMENSIONS

Workpiece length, max.	mm	200
Workpiece O/D, max.	mm	120
Ball section dia., max.	mm	22

MAIN SPINDLE

Spindle flange acc. to DIN 55026	Qty	2
Spindle front bearing diameter	size	6
max. spindle speed	mm	100
Spindle distance	rpm	3000
	mm	500

MAIN DRIVE

AC-Asynchronous motor 60/100 % ED	Qty	2
Max. power	kW	39/28
Full power at spindle speed	kW	39
Max. torque at	rpm	800
	Nm	460

FEED DRIVE

Rapid traverse feed, X	m/min	45
Rapid traverse feed, Z	m/min	30
Feed force, X/Z	kN	11
Ball screw diameter X/Z	mm	50/40

MILLING SPINDLE

AC-motor	kW	15
max. spindle speed	rpm	6.000

ELECTRICAL EQUIPMENT according to VDE 0113

Supply voltage	V	400
Control circuit voltage - DC	V	24
- AC	V	230
Frequency	Hz	50
Total connected load	kVA	70
Input fuse	A	160

DIMENSIONS AND WEIGHT

Length	mm	4250
Width	mm	2650
Height	mm	3000
Weight	kg approx.	12500

Price does not include options (*)

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- 1.5.2 1 SIEMENS OPERATOR PANEL 12" TFT COLOR MONITOR
consisting of OP012 + PCU50 + Control panel (MCP483)
- 1.5.3 1 CNC CONTROL SIEMENS
SINUMERIK 840 Dsl with integrated PLC S7-300
Operator software HMI-Advanced with HMI-Pro
Screen language and on-screen messages in German / English / in official EU
language, more languages available as an optional
CNC program and data memory 3.0 Mbyte, up to 9 MB additional memory is available
as an option.
USB-interface (front) for memory-stick.
Machine diagnostics via internet with VPN technology to verify EMAG's standard format
Software backup is to external media (CompactFlash-Card included and installed in the
PCU50 slot) to verify EMAG's standard format
dataGrabber (Comara) – Capable of recording of process data (raw data) for diagnostics
and process optimization – System is not functional without evaluation software.
Optional tool monitoring:
- Toolinspect (turning, milling), - Zinngrebe (only VLC + grinding)
The control system is designed for a maximum ambient temperature of 45° C
EMAG ADD-ON FUNCTIONS:
Machine-specific documentation of functions and parameters is available directly on the
control unit. (PDF display)
Automatic return to the home position upon machine error. Error recovery is a simple
one button process.
First run high speed traverse is at a reduced speed from home to the first cut. Reduced
speed allows you to avoid a collision and save damaged equipment and reprogramming
costs.
Internal measuring probe with collision monitoring on the traversing movements (Option)
Automatic workpiece gaging when switching to a replacement tool or tool insert (Option)
Automatic filling and emptying of the entire machine.
"Optional Stop" (M01) allows you to make a programmed stop or interrupt production
after finishing a part at a desired point.
Single cycle: Allows one part to be transferred and machined for setup and tryout
purposes.
STRUCTURED WORKPIECE PROGRAMMING:
Division of individual programs into subprograms, e.g. the program for the machining
contains only the technology program and not the loading part.
The loading program has individual parameters. Division into machine and work piece
specific parameters.
Programmable chuck with length measuring system in the pull rod.
Different clamping situations (Unclamped, Clamped and Over stroke) can be changed
by parameter assignment. Displayed by diagnostics menu.
Recording and displaying of the most important production data: work piece counter,
shift counter, cycle time, work piece status.
Production of definable batch sizes. The machine stops upon reaching the selected no.
Optimized programming for the turret keeps chips from collecting.
Rinsing of clamping devices after a programmable number of parts
Number and positions of the workpiece carriers can be defined for closed automation.
Lifecycle screen that displays tool wear and tool life.

Total: 1.119.100,00 €

Price does not include options (*)

Offer : 145205 / 3

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Jürgen Müller, Norbert Heßbrüggen
Handelsregister: Ulm HRB 533796
Steuer-Nr.: 63002/12866 FA Göppingen
Ust-IdNr.: DE 813 898 840

Hyundai WIA Corp.

Bankverbindungen:
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Commerzbank AG Tübingen (BLZ 641 400 36) Konto 8 909 103 IBAN DE09 6414 0036 0890 9103 00 SWIFT Code COBADEFF 641
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**EMAG Maschinenfabrik GmbH**

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E-Mail info@emag.com

EMAG Maschinenfabrik GmbH - Postfach 1152/62 - D-73080 Salach

Commercial Conditions

CONFIDENTIALITY

Commercial and operational data relating to this quotation / order, such as recommended machining data and clamping methods, etc., shall not be divulged to any third party (referring, in particular, but not exclusively, to competitors, for the purpose of bid comparisons) and are to be used solely within the meaning of this quotation / order.

The recipient of this quotation / order expressly accepts these conditions.

CONDITIONS OF SALE

The quotation is subject to the enclosed EMAG conditions of sale.

EMAG accepts therefore, as far as in the enclosed General conditions for sale and delivery not specifically otherwise agreed, no responsibility for indirect or consequential damages, loss in earnings, loss of use, lost production, capital costs or costs for interruption of business.

Claims for damages due to delays in delivery will not be entertained.

EMAG's extended reserve of ownership is considered accepted by the buyer with the issue of a purchase order.

The equipment quoted herein may be subject to German and/or EU export regulations. Therefore, an end user statement may be required for approval by German authorities (BAFA), stating the intended use of the equipment.

MACHINING

The determined cycle times result from values, which are based on data and experiences of acknowledged cutting material manufacturers. Changing rough part dimensions, tolerances, material as well as the material hardness are only possible during building and testing of the machine. Changes will only be accepted in written form and in mutual agreement.

PROVISION OF EQUIPMENT AND MATERIALS BY CUSTOMER

All tooling, workholding equipment, parts, etc. are to be supplied to our works free of charge and carriage paid. We wish to point out that we accept no responsibility for the functional capability and working order of any equipment provided.

The functional capability of the provided equipment is the responsibility of the buyer.

In the event of problems, we assure you of our constructive collaboration. Any additional costs arising from modification, optimization, etc. will be charged to the customer. Furthermore, we do not guarantee any quoted machining times in such cases.

When orders cannot be carried out on schedule, caused by material supplied by the customer or if technical problems should occur, the machine will be invoiced on the originally agreed delivery date, when the relevant payment will become due.

If necessary the agreed upon date of delivery date may be changed to achieve full operability.

It shifts the start of the penalty for delay in delivery accordingly.

DELIVERY PERIOD

Approx. 11-12 months. The delivery period is not binding and can change between the submission of quotation and placement of order.

Please discuss the subject of delivery with us at time of order placement and we will do everything possible to meet your requirements. Changes regarding the component/s or the machines or the order content after order placement may result in changes in price and delivery.

Compliance of delivery date requires timely down and / or partial payment by the customer within the prescribed payment terms. If receipt of down and or partial payment is delayed, the delivery date and the start of penalty for delay in delivery shall be postponed accordingly.

CONFORMITY WITH CUSTOMERS' REGULATIONS

has not been taken into account. Where unavoidable deviations from such regulations prove necessary, specific authorization will be requested.

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PRICES

FCA (Free Carrier). Packing, insurance and transport not included.

The payments are to be made without deduction. With delayed payment the relevant laws apply.

VALIDITY OF PRICES

The prices quoted are valid for 90 days from the date of quotation. Orders placed after this period would be subject to a price increase of 0.5 % per month to cover increases in labour and material costs.

This quotation will replace all previous versions.

PAYMENT TERMS

20 % down payment at order confirmation

70 % at shipping advice (delivery ex manufacturer's works)

10 % at final acceptance in customer's work

All payments due immediately

The payment of a possible contractual penalty is conditional on prompt payments having been made to EMAG. Where payment is delayed, possible contractual penalties will be deferred by the same period.

RETENTION OF TITLE

The machine will remain the property of the builder until complete payment is received from the buyer.

PERIOD OF LIABILITY FOR DEFECTS

12 months at single shift.

The period of liability for defects starts with the final acceptance, but no later than 3 months from delivery, and does not apply to wear and tear parts.

Apart from any possible penalties defined in our terms and conditions, no further claims for damages, consequential damages, etc., will be entertained.

WORKPIECES FOR MACHINE ACCEPTANCE

Both, pre-acceptance and final acceptance tests are carried out with either an EMAG test component to VDI standard, or the workpiece specified by the customer's order.

The customer will supply an agreed number of workpieces for machining process optimisation and preacceptance tests.

These workpieces must be from a single batch and of a quality that corresponds with the relevant manufacturing drawing.

For preparatory work and preliminary machine acceptance the customer will, at the agreed time, supply EMAG free of charge with a sufficient number of sample workpieces of a quality that corresponds with the relevant manufacturing drawing.

Any delay in the supply of these workpieces will result in the machine delivery being delayed and the starting date of a possible delayed machine delivery penalty being deferred accordingly.

FINAL ACCEPTANCE AND FINAL PAYMENT

Final acceptance tests are carried out as soon as the machine has been installed and commissioned.

For this purpose, the buyer has to provide a sufficient number of workpieces, personnel and capacity to measure and evaluate the acceptance components.

Once the machine is taken into production and the agreed component quality and cycle time have been achieved, final acceptance is considered to have taken place.

Where final acceptance of the machine is not possible owing to reasons beyond EMAG's control (missing acceptance components, etc.), final contractual payment falls due immediately.

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STANDARDS AND REGULATIONS

EMAG Machines are built in accordance with the new EC machine guidelines 2006/42/EG and follow the standards and safety regulations. The machine carries the EC-sign that agrees with the guidelines and receives the EC conformity.

TECHNICAL DEVELOPMENT

In the interest of technical progress, we reserve ourselves the right to make design changes which are due to continuous product development.

RESERVATION CLAUSE

This offer is subject to the control and to the granting of an export license by the German Federal Office of Economics and Export Control (BAFA).

Yours faithfully,

EMAG Maschinenfabrik GmbH

Bernhard Böttigheimer
Managing Director

Hartmut von Fugler

Enclosures

EMAG-General conditions for sales and delivery (UN)
Layout
Timestudy
Schematic of machining

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